Linear Regression as an Intuitive Design Tool for Optimal Parallel Elastic Actuators

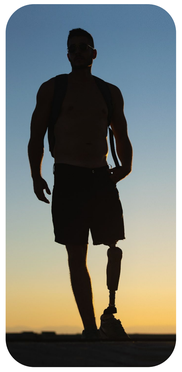
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Background

- There are currently **2.1 million** people in the US living with **limb loss**
- Globally there are over 1 million amputations each year [1]
- **70%** of prosthetic devices in use are for **below-the-knee** amputations [1]



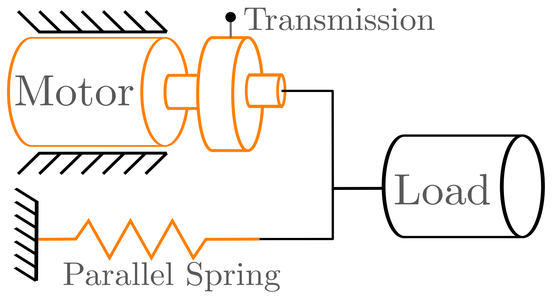


A render of the University of Michigan's OSL [2]

- Our lab specializes in **compliant mechanisms**, which **bend and deform** to store/release energy
- These are **significant advantages** for powered and passive **prosthesis**

Parallel Elastic Actuators

- **Parallel Elastic Actuators** (PEAs): any system with an elastic element (springs) in parallel with a powered actuator (motor)
- PEA's can **reduce power** and load requirements, **extend runtime**, and enable **cheaper actuators**.
- The stiffness is dependent on the **load/deflection profile** for the task
- There is a range of stiffnesses that are beneficial



Regression

Summary

- **Optimal stiffness** for a PEA can obtained by **minimizing** an objective, such as **RMS Torque**
- Usually computed via convex optimization [3]
- We discovered **Weighted Least Squares Regression** (WLSR) is an alternate approach
  - Produces a **single optimal** spring
  - Provides **visual intuition**
  - Maintains a **simple** mathematical formulation
  - Enables the design of **higher-complexity** elastic elements, such as **piecewise springs**

Objective

$$min: \sum_{i=1}^{n} w_i (y_i - \tau_i)^2$$

(RMS Torque Minimization as Weighted Regression)

System Inputs

Task Definition (y_i = *walking, stairs*)

Actuator Dynamics $J_m, \mu, \eta, k_m, k_t, \Omega$

Preprocessing

Discretize trajectory ($i = 1 \dots n$)

Weight by duration ($w_i = \Delta t_i$)

Select stiffness model (τ_i = *piecewise, polynomial*)

Output

Unique Optimal Stiffness Coefficients

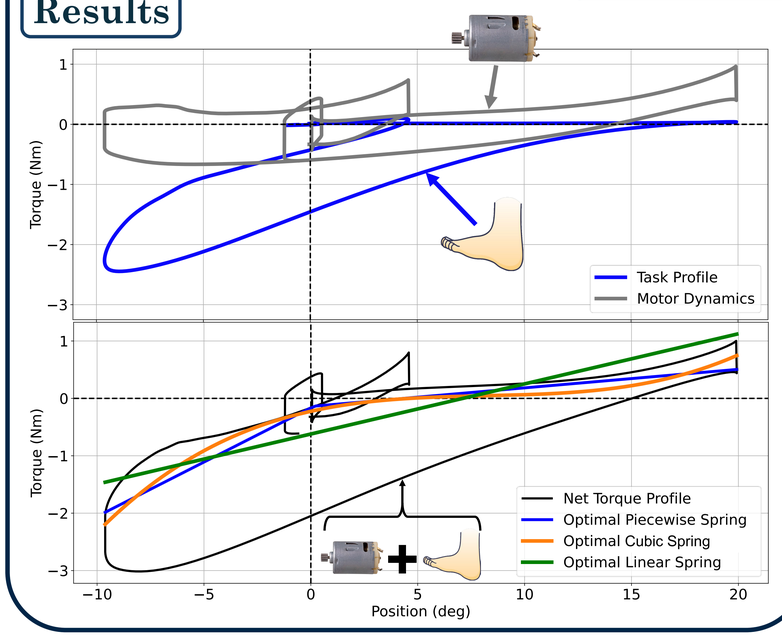
Piecewise Linear Stiffness

$$\tau_s = \begin{cases} A_1 x + B_1; x \le C \\ A_2 x + B_2; x > C \end{cases}$$

Polynomial Stiffness

$$\tau_s = A_n x^{n-1} + \cdots A_1$$

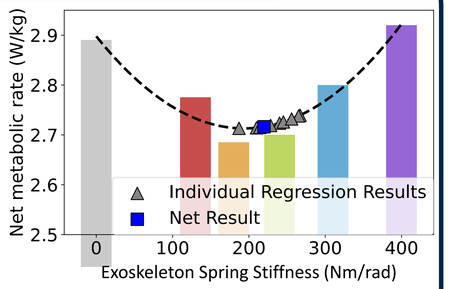
Results



Validation

External Dataset

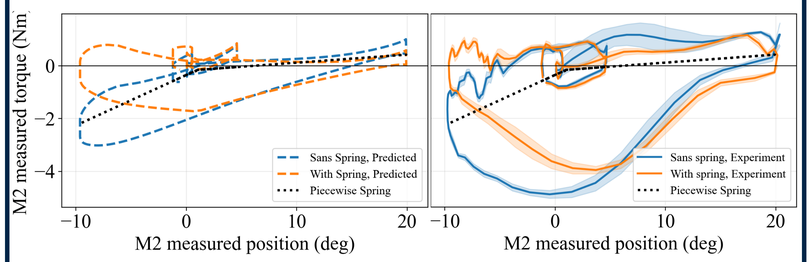
- Ankle walking data from Collins et al. [4] was used to **evaluate** WLSR on a **different dataset**
- WLSR reported an overall stiffness of **219** Nm/rad, within the **180** and **240** Nm/rad reported as **metabolically optimal**
- Subject-specific regressions yield spring constants in the range of **187**-**267** Nm/rad, indicating that **WLSR remains effective** for alternate datasets



Piecewise Spring

- **Benchtop validation** produced a walking profile **via motors**, with an optimal **piecewise spring**.
- **Predicted RMS Torque** reduction was **46%**
- **Measured RMS Torque** reduction was **30%***





*Predicted benefit using the Experimental data and piecewise spring model was **14%** reduction in RMS Torque

Sources

[1] Pro Medical East. "29 Limb Loss Statistics, Facts & Demographics." *Pro Medical East,* 31 Mar. 2023, promedeast.com/limb-loss-statistics/.
[2] opensourceleg.org. Accessed 26 Aug. 2026.
All uncaptioned images are our own or sourced from Adobe Stock (Educational License).
[3] K. Yang, M. Dickens, J. Schmiedeler, and E. Bolivar-Nieto, "Extending the benefits of parallel elasticity across multiple actuation tasks: A geometric and optimization-based approach," pp. 1–11
[4] S. H. Collins, M. B. Wiggin, and G. S. Sawicki, "Reducing the energy cost of human walking using an unpowered exoskeleton," vol. 522, no. 7555 pp. 212–215.

Acknowledgements We acknowledge funding from the National Science Foundation under Grant 2344766. This work is in collaboration with the University of Notre Dame.